FRESHSENSE INNOVATIONS

EST. 2025

"Freshness at a glance."

START-UP BUSINESS PLAN

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03/4/2025

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Executive Summary

Overview

FreshSense Innovations is an innovative food safety solutions company which was designed to ensure the freshness of the food items. FreshSense containers are smart food containers which provide a simple indication to detect food freshness. The storage container lids have built-in pigment sensors that react to harmful gases and odors to detect whether your food has gone bad. The food storage container is durable, provides real-time freshness monitoring, and protected pigments to detect spoilage. The container lid transitions between green(fresh), yellow(aging), and red(spoiled) to make sure that your food is safe to eat. FreshSense address the global food waste concerns by providing consumers with better tools to help address food spoilage. The Freshness container comes in different sizes to meet the needs of the consumer. The Freshness container reduces food waste and ensures food freshness for all.

Unique Value Proposition

FreshSense is revolutionary food storage product that prioritizes food safety and personal health. The innovative color-changing lid technology and real-time monitoring, it ensures to give updates about food freshness. It provides clear visual signals through safe pigment indicators and promotes healthy living. The container is made out of glass, not plastic which makes it more durable and longer lasting to fit every customer's need.

Key Metrics

Market Penetration

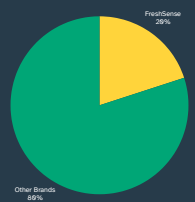
- Achieve 20% market penetration within the first three years

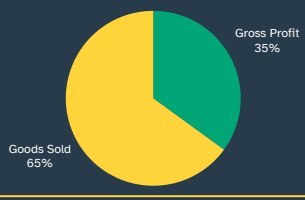
Customer Retention

- Achieve a 70% customer retention rate by the second year

Gross Profit Margin

- Maintain a 35% gross profit margin





Customer Segment

Primary Market	Secondary Market
FreshSense targets people who want to prioritize their health and are environmentally conscious. This also includes restaurants and other food services who want to ensure a safe way to store food.	Partnerships with local grocery stores, cafes, and wellness centers that want to offer high-quality and nutritious products to their customers.

Problems

Global Food Waste: Almost 1.3 billion of tons of food are wasted each year due to spoilage and lack of proper storage. This is a waste of food and resources, which can also contribute to significant economic problems.

Health Risks: Millions of illnesses such as E. coli and Salmonellosis come from consuming spoiled food each year. This can lead to dangerous health consequences long-term, and extreme healthcare costs.

Lack of Reliable Products: Many traditional food storage companies do not provide tools to detect food freshness. This forces consumers to find food spoilage through unreliable methods such as smell.

Solutions

Monitor Freshness Real-Time: The odor and gas sensitive pigments change color to provide a visual indicator to determine the status of the food. This allows the consumers to act before the food becomes spoiled.

Safe Spoilage Detection: The Freshness food storage container removes the need for dangerous practices such as tasting questionable food. The pigments in the container allows for visual freshness indicators, reducing the risk of illnesses.

Reliability: The pigments in the product allow the consumer to visually see the freshness of the food. This removes unreliable methods such as smell or appearance, and replaces it with color-coded freshness.

Channels

Retailers

E-Commerce

Social Media

Conclusion:

FreshSense is seeking a $600,000 initial investment in return for 8% equity and plans to disrupt the $2 billion global food storage industry. The funding will enable the company to scale its manufacturing ability, expand its marketing efforts, and continue to strengthen its advantage with the innovative FreshSense design. Investors should expect a return of 18%. The investment will be paid back in full in three years, and investors will receive 2% per year based on their investment as a dividend. Through its unique approach, FreshSense is well-positioned for long-term sustainable growth and profitability.

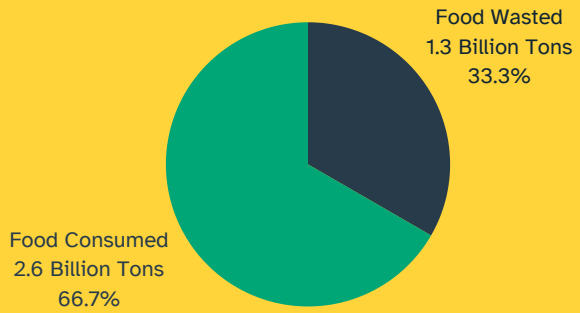
PROBLEMS

WHAT DO THE FRESHSENSE CONTAINERS SOLVE?

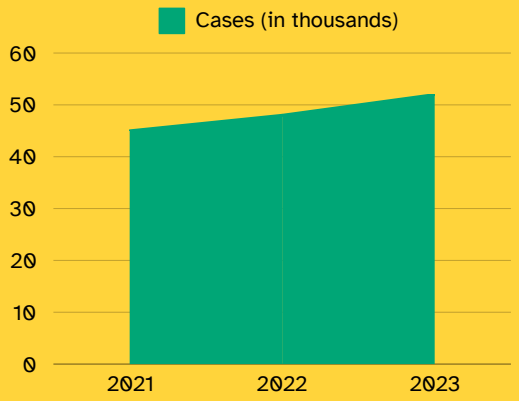
PROBLEM #1: FOOD SPOILAGE AND WASTE

Over $1 trillion per year are spent globally due to food waste, according to the United Nations Food and Agriculture Organization (FAO). Making food at home definitely comes with a cost. Without expiration dates, like the ones on packaged food, estimating when the leftovers will go bad is very difficult. Containers in today's day and age are not always helpful in preventing the perishing of our food and do not do a good job in notifying us when it has. When stored for long periods of time, the food is inevitable to go bad, and is eventually thrown out. Roughly 1.3 billion tons of food are wasted annually, making up one-third of global food production. People often neglect leftovers, telling themselves that they will eat it eventually, and most of the time, this never happens. Ultimately, reducing food waste is a huge factor in improving sustainability, increasing resource allocative efficiency, and lowering hunger rates.

GLOBAL FOOD PRODUCTION ANNUALLY



Food-borne Illnesses in the European Union

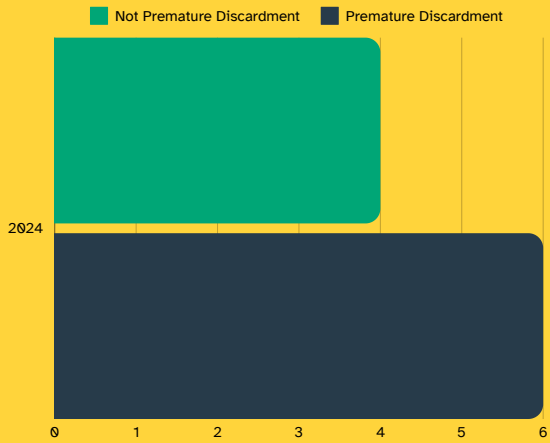


PROBLEM #2: HEALTH RISKS

The leading impact of spoiled food is food-borne illnesses, which, according to the World Health Organization (WHO), affect over 600 million people worldwide every year. Those diseases kill 420,000 people a year, hitting hardest the most vulnerable, including young children, pregnant people, and the elderly. Toxic "invisible" spoilage can occur from bacteria like Clostridium botulinum, Listeria monocytogenes, Salmonella, and Escherichia coli (E. coli), which quietly produce toxins from within food. Its appearance, smell, and taste often remain unaffected, leading to the "invisible" effect. However, the signs when food is bad aren't apparent to everyone, and some people unwittingly eat food that is bad. Not all tools for assessing food safety are reliable and easy to use, and that has led to guesswork about whether food is safe to consume. This elevates the risk of disease and adds to high treatment costs, estimated at $15 billion a year in the United States alone.

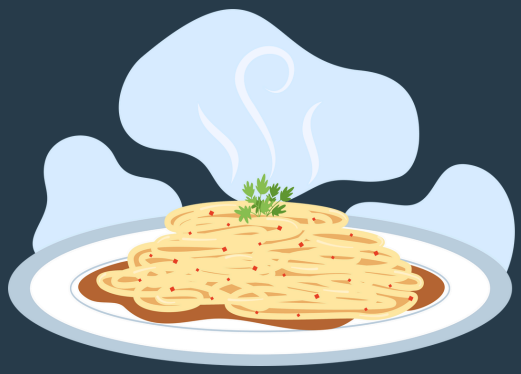
PROBLEM #3: LACK OF RELIABLE INDICATORS

Traditional methods of determining whether food has gone bad are not always the safest options. Usually, when spoiled food is indicated, it's already too late. Standard food containers lack mechanisms to indicate and alert users of spoilage, often leading to guessing. Electronic freshness indicators are rare on the market, and the ones that are available are expensive and easy to damage. Most people end up smelling the food, or sometimes even tasting it, which can result in illness, and in some extreme cases, death. A study by the National Institute of Food Technology found that 60% of consumers have discarded food prematurely due to uncertainty about spoilage, even when the food may still be safe to eat. Often, people simply discard the food, out of caution of it being spoiled, which contributes to even more unnecessary food waste. All in all, without efficient indicators, people end up with health complications, or simply wasting the food.



CUSTOMER SEGMENTS

WHO WILL USE FRESHSENSE CONTAINERS?



MARKET SEGMENTATION

The target demographic for the FreshSense Containers includes all ages with families, urban professionals, and middle-income households, and those who are eco-conscious. These consumers are health-conscious and seek fresh, safe food, as well as environmentally aware individuals who are motivated to reduce global food waste. According to the USDA, the average American family of four wastes an estimated 1,500 dollars worth of food in just one year. Left overs are inevitable, and FreshSense Containers increase the efficiency of food storage by ensuring the safety of the food stored. Geographically, the initial focus will be on high-population areas such as New York, Texas, Florida, and California, followed by expansion to regions with sustainable, eco-conscious living. The ultimate goal is global expansion, particularly in regions like Canada and the United Kingdom, where freshness and sustainability are prioritized.

SECONDARY MARKET

Our secondary market partners with food retailers, fresh food transport businesses, and companies with strong eco initiatives to reduce food waste. We work with a wide range of businesses, from small restaurants, grocery stores, and food delivery services to big players in the food distribution industry. Alarmingly, around 84.3% of unused food in restaurants is thrown away, with only 14% being recycled and a mere 1.4% donated, according to *Fourth*. FreshSense helps them provide innovative solutions for extending shelf life for fresh and pre-cooked products. Our plan is to integrate our solutions into their operations to reduce their environmental impacts, promote sustainable practices, and drive a larger transformation toward responsible food consumption.

PRIMARY MARKET

Our target market is a health-conscious, eco-conscious set of individuals in the market looking for fresh food container solutions to reduce food waste. FreshSense's products focus on both freshness and sustainability, enabling it to become an indispensable tool for families and busy urban professionals. According to a 2023 report by the National Frozen and Refrigerated Foods Association, 81% of U.S. consumers cook more than half of their meals at home. As this comes out, people are yet to become aware of the environmental implications for food waste and are still searching out convenient, safe ways of reduction. At FreshSense, the mission is about empowering the customer through innovative solutions which drive sustainability by a reduction in waste, with the quality of each meal retained. Available everywhere, from a wide network of retailers and directly on our website, FreshSense wants to make life easy for customers looking to make environmentally responsible purchasing choices. The intelligent choice: FreshSense is available in stores and online.

UNIQUE VALUE PROPOSITION

WHAT MAKES FRESHSENSE CONTAINERS SPECIAL?

Simplifying, Improving, and Saving Lives.

Eco-Friendly

Durable and Dishwasher-Safe

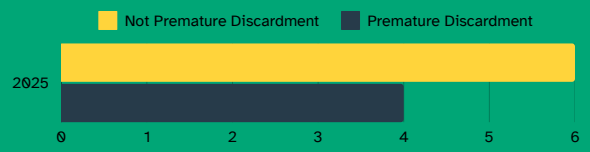
Food-Safe Biochemical Pigments for Instant Freshness Status

SOLUTIONS

HOW DO THE FRESHSENSE CONTAINERS SOLVE THESE PROBLEMS?

SOLUTION #3: RELIABLE TECHNIQUES

The FreshSense containers provide reliable storage solutions that other traditional brands do not. Many popular brands such as Rubbermaid and Snapware do not have tools to detect food freshness, and leave consumers to have to use unreliable methods such as smell or appearance. The FreshSense containers are more reliable due to the pigments used to provide visual indication for food freshness. The built-in color coded pigments are accurate indicators that show the quality of the food, which eliminates the uncertainty and ensures that users can determine if their food is safe to eat. FreshSense offers a trustworthy and innovative solution to ensure that the consumer is confident in what they are eating is safe.

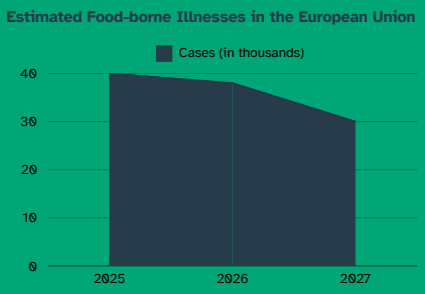


SOLUTION #2: SAFE SPOILAGE DETECTION

The FreshSense detection provides safer practices for identifying spoiled food. The FreshSense container provides visual detection of the freshness of the product, which eliminates the need for potentially dangerous practices such as smelling or tasting the questionable food. The visual indicators reduce the risk of consuming spoiled food and directly address the health risks. Many food-borne illnesses such as E. Coli and Salmonellosis cause dangers to millions of people every year. The FreshSense containers promote safer storage and consumption of food. Dangerous practices such as smelling or tasting can have severe health effects and can be very unreliable. The FreshSense containers minimize food-borne illnesses, and ensures better health outcomes for the consumers.

SOLUTION #1: MONITOR FRESHNESS IN REAL-TIME

At the forefront, the FreshSense container employs innovative gas-sensitive pigments embedded within the lids that actively scan the internal environment for spoilage indicators. These pigments, including methylene blue, can blue, and phenolphthalein, respond to gases like hydrogen sulfide, ammonia, and carbon dioxide, which are produced as food decomposes. As the food evolves, the pigments shift through a glossy, visual rainbow of colors: from green (denoting freshness), to yellow (indicating aging), to red (showing rot). Once the spoilage stimuli are eliminated, for instance, if the food is air-exposed or properly packaged, the color pigments will revert to their natural color. They even claim that their pigments are designed to reset after a wash in the dishwasher. This color-changing technology in real time enables consumers to better gauge food-refrigerated conditions to help eliminate unnecessary waste and promote better food storage.



CHANNELS

WHERE CAN YOU PURCHASE FRESHSENSE CONTAINERS?

LOOK FOR THIS BOX!

E-COMMERCE/WEBSITE

FRESHSENSE.CO

Convenience to our customers is most important at FreshSense, with our official website representing the best avenue for learning about our FreshSense Containers, accessing special promotions, and using our "Get In Touch" service, through which they can effortlessly communicate via email or by phone. Placing an order on our website is effortless, taking only seconds to find your new container, add it to cart, and place the order! Customers can be assured of purchasing with convenience, flexibility, and yet rich in services and accessibility from us. Positive reviews left under the product description allows for us to improve our products and encourage people to consider trying out FreshSense's innovative selection.

SOCIAL MEDIA

TIKTOK, SNAPCHAT, INSTAGRAM, FACEBOOK, YOUTUBE

Through social media's growing commerce platforms such as Tiktok Shop, Instagram Shops, Facebook Marketplace, etc., customers can easily place orders for our products. Health and culinary influencers receive affiliate benefits for promoting our products by making demonstration videos on Tiktok, Snapchat, Facebook, Youtube, Instagram, etc. Links on advertisements bring in organic traffic to our website, and can increase customers likelihood of purchasing a FreshSense product by up to 75%.

DEMOGRAPHICS

- 88% of consumers research products online before buying (Oberlo, 2025).
- 76% of shoppers prioritize convenience when choosing brands (PwC, 2025).
- 95% of buyers read reviews, and positive reviews boost purchase likelihood by 270% (BrightLocal, 2025).
- 70% of mobile users complete purchases in under 5 minutes with seamless checkout (Statista, 2025).

- 5.8 billion users in 2025 spend an average of 2.5 hours daily on platforms (DataReportal, 2025).
- Ad spending will exceed $226 billion, with 93% of marketers citing social media as best for ROI (Statista, 2025).
- 83% of Instagram users discover new products (Hootsuite, 2025).
- TikTok's ad revenue is expected to reach $15 billion (Hootsuite, 2025).
- Influencer marketing is now a $32.5 billion industry (Influencer Marketing Hub, 2025).
- 48% more people are watching video content, showing shifting consumer behavior (Influencer Marketing Hub, 2025).

RETAILERS THAT SELL FRESHSENSE CONTAINERS:

Walmart	**Look for our containers in the "Food Storage" aisles.**
Target	**Look for our containers in the "Kitchenware" aisles.**
Costco Wholesale	**Look for our containers in the "Housewares" aisles.**
Amazon/Amazon Fresh	**Look for our containers in the "Storage" aisles/sections.**

WHY RETAIL?

New customers who are not familiar with FreshSense can discover us in retail stores. Most retailers will have one of our containers out on display, giving the customer a hands on look and feel of the product. Our trusted retailers allow FreshSense to grow its customer base while maintaining existing purchasing options. According to the National Retail Federation (2020), selling in retail stores boosts a company's presence, with 70% of consumers saying they are more likely to trust a brand that has a physical store location. Our company has proven this statistic to be valid as our sales have grown over 60% through retailers.

REVENUE STREAMS

HOW DO WE BRING IN REVENUE?

LIFETIME VALUE

Lifetime Value helps FreshSense assess the revenue generated by a single customer over their lifetime, and provides insights into long-term profitability and retention rates. For one-time customers, the estimated average spend is $110, factoring in upselling and potential future product offerings. The approximate LTV for this group is $110 per customer. Key revenue sources include container packs available in 1-pack, 3-pack, 5-pack, and 8-pack options, which are made to help satisfy consumers diverse needs, as well as encouraging bulk purchases. By offering high-quality, reusable containers, FreshSense creates customer satisfaction and loyalty, which ensures consistent revenue growth and builds consumer trust over time.

OVERVIEW

FreshSense Innovations create a wide array of revenue streams to meet the needs of health and eco-conscious consumers. These are categorized into sales channels: direct-to-consumer online and retail partnerships, with product offerings. Our biochemical technology-driven freshness monitors reset automatically, designed to minimize waste while maximizing convenience. This simplicity ensures consistent product quality and affordability, positioning us as a leader in food freshness and sustainability.

Revenue Streams	Price	Unit Sold	Sales Revenue
1 - Pack	$30.00	125,000 units	$3,750,000
3 - Pack	$85.00 ($28.33 per unit)	60,000 bundles (180,000 units)	$5,100,000
5 - Pack	$135.00 ($27.00 per unit)	40,000 bundles (200,000 units)	$5,400,00
8 - Pack	$200.00 ($25.00 per unit)	30,000 bundles (240,000 units)	$6,000,000
Total			$20,250,000

PACK OPTIONS

1-Pack
- Price: $30.00 per pack
- Unit Cost: $19.50 per pack
- Includes: 1 freshness container
- Total Sales Revenue: $3,750,090

3-Pack
- Price: $85.00 per pack ($28.33 per unit)
- Unit Cost: $58.50 per pack
- Includes: 3 freshness containers
- Total Sales Revenue: $5,190,090

5-Pack
- Price: $135.00 per pack ($27.00 per unit)
- Unit Cost: $97.50 per pack
- Includes: 5 freshness containers
- Total Sales Revenue: $5,400,090

8-Pack
- Price: $209.00 per pack ($25.00 per unit)
- Unit Cost: $156.00 per pack
- Includes: 8 freshness containers
- Total Sales Revenue: $6,000,090

COST STRUCTURE

WHO WILL USE FRESHSENSE CONTAINERS?

DISTRIBUTION COSTS

Annual Distribution Costs

- E-commerce Shipping:
  - $5.00 per unit (packaging & shipping for online orders).
- Retail Distribution:
  - $100,000 (stocking fees for Walmart, Target, Costco, Amazon Fresh).
- Warehousing:
  - $50,000 (storage & inventory management).
- Total Annual Distribution Cost
  - 745,000 units × 3,875,000 * * *(745,000 units × 5) + 100,000 + 100,000 + 50,000 = **$3,875,000**

CAC (CUSTOMER ACQUISITION COST) + MARKETING COSTS

Marketing & Customer Acquisition Cost (CAC)
Total Annual Marketing Budget: $300,000

- Digital Marketing: $150,000 (social media ads, influencers, Google Ads).
- Retail Marketing: $100,000 (in-store promotions, partnerships).
- Content Creation: $50,000 (videos, blogs, testimonials).

Customer Acquisition Cost (CAC): $15 per customer

- Breakdown: 300,000(total marketing cost) ÷ 20,000(new customers) = 300,000(total marketing cost) ÷ 20,000(new customers) = 15 CAC.
- Justification: Sustainable with a projected LTV of $1,100 (CAC:LTV = 1:73).

MANUFACTURING COSTS

1. Material Costs
   - Glass Containers: $10.00/unit
   - Pigment pigments: $2.00/unit
   - Lid Mechanism: $5.00/unit
     - includes encapsulation and integration of pigments
   - Other Materials: $2.50/unit
     - per unit packaging, labels, etc.
   - Total Material Cost $19.50/unit
2. Equipment & Facility Costs
   - Machinery for Glass Molding: $100,000/year
   - Pigment Integration Equipment: $50,000/year
   - Facility Rental: $50,000/year
3. R&D Costs
   - Pigment Research & Testing: $100,000/year

Total Manufacturing Costs: $300,000/year

HUMAN RESOURCES COST

- Recruitment Expenses: $25,000
  - Job postings on platforms like LinkedIn and Indeed.
  - Agency fees for hiring specialized roles (e.g., R&D scientists, marketing leads).
  - Background checks and onboarding paperwork.
- Training & Development: $20,000
  - Onboarding programs for new hires.
  - Workshops on sustainability, food safety, and product innovation.
  - Leadership development programs for future managers.
- Salaries for HR Staff: $60,000
  - Compensation for HR managers and support staff.
  - Payroll processing and administrative support.
- Employee Benefits: $50,000
  - Health insurance contributions (medical, dental, vision).
  - Retirement plan matching (e.g., 401(k) contributions.
- Performance Incentives: $15,000
  - Bonuses for employees who meet or exceed performance goals.
- Compliance & Tools: $10,000
  - HR software (e.g., payroll systems, employee management tools).
  - Legal compliance (e.g., labor laws, workplace safety regulations).

Total Annual HR Costs: $180,000

KEY METRICS

HOW WILL WE MEASURE OUT GROWTH?

MARKET PENETRATION

- **Market Penetration:**
  - A key metric used to determine the percentage of the total market that a product has captured. It can be measured by the number of customers and sales relative to the market size
- **How will we achieve our goal?**
  - FreshSense is willing to invest $500,000 into marketing strategies such as targeted ads and SEO to expose the brand
  - The company will partner with online platforms such as Amazon and secure retail partnerships by Year 2
  - The money will contribute towards sponsorships will influencers who will help get the name in the market
- **Market Penetration by Year:**
  - Year 1 --> 5% Market Penetration
  - Year 2 --> 12% Market Penetration
  - Year 3 --> 20% Market Penetration

CUSTOMER RETENTION

- **Customer Retention Rate:**
  - A key metric that measures the percentage of customers that continue to use a company's product over a given period. It can be measured through dividing the number of customers at the end by the amount at the start, and subtracting the new customers during the period.
- **How will we achieve our goal?**
  - FreshSense will launch loyalty programs that include perks such as free shipping or discounts to ensure that customers continue to use out products
  - There will be designated emails for frequent or dormant buyers to send tailored offers
  - Customers will have access to frequent customer service centers with trained staff to provide quality service for every customer

GROSS PROFIT MARGIN

- **Gross Profit Margin:**
  - A key metric that measures profitability by calculating the percentage of revenue remaining after the cost of goods sold.
  - FreshSense will have $0.35 in gross profit after production costs for every $1 in revenue, which will maintain a 35% gross profit margin.
- **How will we achieve our goal?**
  - FreshSense will optimize production costs by using bulk manufacturing and cost-effective materials to reduce the COGS
  - Offer different price points for bundles and multi-pack purchases
  - Increase sales channels through e-commerce, retail partnerships, and DTC websites

COMPETITIVE ADVANTAGE

HOW IS FRESHSENSE BETTER THAN EXISTING PRODUCTS?

WHAT MAKES US UNIQUE?

FreshSense differentiates itself due to its innovative consumer freshness monitoring. The FreshSense technology revolutionizes how individuals and businesses track the freshness of perishable goods. The FreshSense container does not rely on traditional methods such as expiration dates, and instead used reliable, real-time freshness indicators. FreshSense is able to stand out in the highly competitive market with its smart pigment technology and data-driven insights. The unique positioning lowers the chances of new products, and allows FreshSense to establish a distinct position within the industry. FreshSense operates with a scalable production model, so when demand increase, the cost per unit decreased. This will help strengthen out competitive advantage, allowing for future market expansion. Our commitment to sustainability and reducing food waste aligns with global initiatives and fostering partnerships. These strategic partnerships are laying the foundation of lasting influence and stable growth.

Key Differentiators	FreshSense	Rubbermaid	Snapware	FineDine
Real-Time Freshness Detection	✓	✗	✗	✗
Help Prevents Food Waste	✓	✗	✗	✗
Durability	High	High	High	Medium
Reliability	High	Medium	Medium	Low

CONCLUSION

WHY SHOULD YOU CHOOSE US?

WHY SHOULD YOU INVEST?

FreshSense is seeking a $600,000 investment for 8% equity with the mission to reduce the billions of dollars wasted in the global food industry. This funding will fuel research and development, expand production, and enhance the overall FreshSense product to ensure that FreshSense remains at the forefront of innovative food storage. Investors can expect their full investment to be paid back in three years, along with an 18% expected return and 2% annual dividend based on their initial investment. This return provides short-term and long-term gains so FreshSense can cement itself as as industry leader.

ALLOCATION OF FUNDS

- $200,000 - Expanding Manufacturing Capabilities to Support Large-Scale Production
- $150,000 - Strengthening Market Efforts to Increase Market Penetration
- $150,000 - Advancing R&D to Improve FreshSense Products to Maintain Competitive Differential
- $100,000 - Build strategic partnerships with retailers and distributors to drive widespread adoption

MONTHLY PAYMENTS BREAKDOWN

Total Return on Investment	$108,000
Total Amount Paid	$708,000
Monthly Payment of Principal	$16,666.67
Monthly Interest Payment	$2,000
Total Monthly Payment	$18,666.67

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